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Heat source models for numerical simulation of laser welding processes – a short review

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Abstract. In recent decades, numerical modeling and computer simulation have become an integral part of the design, analysis and optimization of fusion welding processes, including laser welding. In general, laser welding processes involve the interaction of multiple physical phenomena, such as thermal, fluid, metallurgical, chemical, mechanical, and diffusion effects, which makes the development of a simulation model difficult and complex. In addition to the geometric characteristics of the parts to be welded, their material properties must be specified in a wide temperature range, as well as the conditions for heat removal to the environment or shielding gas. One of the most complex tasks in the preparation of a simulation model of the laser welding processes consists in the selection of an appropriate heat source model to accurately determine the heat input to the weld. Very important is also the process of experimental verification and validation of the developed simulation models. In this paper, a short examination of significant mathematical heat source models for numerical simulation of laser welding is provided. Numerical analysis of laser welding of sheets made of S650MC steel is accomplished using conical 3D heat source model with the support of the ANSYS code. The effect of geometrical characteristics of the conical volumetric heat source model on the computed width, length, and depth of the weld pool is discussed, along with evaluation of maximum weld pool temperature.

1. Introduction

Modelling and numerical simulation of laser welding processes is currently widely used not only to design and optimize welding parameters but also to predict final properties of the weld joints, their distortions, microstructure, phase composition and generally the weld quality [1-6]. It significantly contributes to a deeper understanding of the underlying physical phenomena associated with welding processes and their mutual interaction.

Progress in the field of modeling and numerical simulation of welding processes is tightly connected to the continual evolution and advances in computational methods and computer capabilities. From the simple solution of temperature fields, the modeling of welding processes has now moved into the area of solving complex coupled problems, which include the analysis of temperature, fluid, stress-strain, electro-magnetic and diffusion fields taking into account metallurgical transformations [7-12].

The primary challenge in numerical simulation of welding and related processes lies in modeling the heat input for the thermal analysis. The mathematical models provide a way to estimate the heat input during different welding processes. Understanding and controlling the heat input is critical for achieving the real and sufficiently accurate results of numerical simulation of welding processes [13-17].



Typically, the heat input to the weld is based on the concept of a heat source - point, linear, surface or volumetric. The selection of a suitable heat source model is usually followed by calibration and optimization of its parameters [18–27].

The paper presents a short overview of heat source models applied primarily to the modeling of heat input in laser welding processes.

2. Mathematical modelling of temperature fields in fusion welding processes

In the context of welding, the transient temperature fields can be mathematically described using the heat conduction equation - Fourier-Kirchhoff's partial differential equation in the form [28]

$$\rho c_p \frac{DT}{Dt} = \left[\frac{\partial}{\partial x} \left(\lambda_x \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(\lambda_y \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(\lambda_z \frac{\partial T}{\partial z} \right) \right] + q_v. \quad [\text{W.m}^{-3}] \quad (1)$$

This equation incorporates a term DT/Dt which denotes substantial derivative of temperature T [$^{\circ}\text{C}$, K] with time t [s], material properties: density (ρ [kg.m^{-3}]), specific heat (c [$\text{J.kg}^{-1}.\text{K}^{-1}$]), thermal conductivity tensor (λ [$\text{W.m}^{-1}.\text{K}^{-1}$]), and volumetric density of internal heat sources (q_v [W.m^{-3}]). This term, representing the heat generated in the unit material volume per unit time, can be used for modelling of a heat source in welding processes.

To solve the heat conduction equation (1) using the theory of partial differential equations, it is necessary to establish uniqueness conditions, which means that several crucial inputs must be provided:

- *Geometric Parameters* - define the shape and dimensions of the body in which the temperature field is calculated. They are necessary to establish the spatial characteristics of the welded structures.
- *Material Properties* - thermo-physical properties of material(s): density (ρ), specific heat (c), thermal conductivity (λ), and other relevant material characteristics. This refers to the material properties of all the base material(s) and possibly the filler material used in welding. When entering material properties for fusion welding processes, it is necessary to take into account their dependence on temperature.
- *Initial Conditions* - describe the initial temperature distribution at the beginning of a process in time $t = 0$ second, i. e. at the beginning of the welding process. They serve as the starting point for the mathematical analysis and influence how temperature evolves over time.
- *Boundary Conditions* - describe the thermal behavior at the boundaries or interfaces of a body (system) and specify how heat is exchanged with the surrounding environment. In welding processes, they specify the thermal interactions at the interfaces of the welded structures and their surroundings. They play a crucial role in determining how heat flows in and out of the welded region, affecting the welding process and the resulting temperature fields.

Mathematically, various kind of boundary conditions (BC) can be defined. The boundary condition of the 1st kind (Dirichlet BC) is given by the body surface temperature. The boundary condition of the 2nd kind (Neumann BC) specifies the conductive heat flux on outer body surface (on the outer surface welded structures). This condition can be also used to define a surface heat source in welding processes. The boundary condition of the 3rd type (Fourier BC) represents the equality between conductive and convective heat flux on the outer body surface. This condition is determined by the ambient temperature and the heat transfer coefficient.

Generally, the boundary conditions can vary with temperature or over time. Combined boundary conditions frequently arise, particularly when heat exchange between a system and its surroundings is facilitated through both convection and radiation. This is also the case for components cooling in welding processes. The total (combined) heat transfer coefficient, denoted as h_C , is calculated as the sum of the convective heat transfer coefficient (h_K) and the reduced (or equivalent) radiation heat transfer coefficient (h_R)

$$h = h_C + h_R = h_C + \frac{\varepsilon \sigma_0 (T_w^4 - T_f^4)}{T_w - T_f}. \quad [\text{W.m}^{-2}] \quad (2)$$

where ε [-] is the emissivity and σ_0 [$\text{W.m}^{-2}.\text{K}^{-4}$] is the Stefan-Boltzmann constant, T_w and T_f are the surface and fluid (ambient) temperatures, respectively. The values of the convective heat transfer coefficient can be calculated using appropriate correlations for natural or forced convection [28].

The approach to defining loads in welding processes is based on the selection and specification of a suitable heat source model describing the heat input to the weld. It is necessary to note that the choice of the heat source model depends on the specific welding process, materials, and welding parameters being used. This procedure is one of the most difficult parts in developing a simulation model of the welding processes. On the other hand, the accuracy of the applied heat source model is crucial for reliable simulations and predictions of the temperature and thermal behavior of structures during welding processes.

3. Heat source models in modelling of laser welding processes

The history of heat source models in numerical simulations is closely connected to the development of numerical methods and computational tools for solving complex thermal and fluid dynamics problems. A brief overview of the development of heat source models in numerical simulations of laser welding processes involves a simple point heat source model, some surface heat source models and mainly different types of volumetric heat source models.

3.1. Simple point heat source model

In 1941, David Rosenthal [29] presented an analytical solution for the heat transfer equation, taking into account the effect of a point heat source moving on the surface of a semi-infinite body. Even today, this solution is still employed for a quick and efficient first approximation of temperature fields during welding processes, despite its reliance on a number of simplifying assumptions. These assumptions include constant thermo-physical properties, the absence of phase change, no heat transfer by convection and radiation from the body to the surroundings (adiabatic conditions), and a quasi-steady state temperature distribution. Moreover, a point heat source exhibited an infinite heat input (singularity) in close proximity to the source point. The solution of the Rosenthal equation for the heat source moving in x -direction is [29]

$$T(r) = T_0 + \frac{\Phi}{2\pi\lambda r} \exp\left[-\frac{c\rho w(r+x)}{2\lambda}\right]. \quad [^\circ\text{C}] \quad (3)$$

where r is the radial distance from the origin location [m], $T(r)$ is the temperature at the radial distance, T_0 is the initial temperature of semi-solid body [$^\circ\text{C}$], w is the velocity of a moving heat source [m.s^{-1}], and Φ is the heat flow [W] representing the effective power of a heat source.

Several authors accomplished analytical solution of heat conduction equation with some modifications of proposed solution conditions [30-33]. A comprehensive summary of analytical solutions for the heat conduction equation with moving point heat sources is available in the handbook published by Carslaw and Jaeger [34] or in the work by Volcenko et al. [35].

The disadvantage of these heat source models is that they ignore the energy distribution in the real heat source. These models describe the energy concentrated in one point or segment, which means neglecting the shape of the weld pool and the depth of penetration [15, 31, 36].

3.2. Surface heat source models

In the case of surface models of the heat source, it is assumed that they act on a defined part of the surface area of the welded constructions. They are incorporated into the simulation model using the boundary condition of the second kind, i. e. through heat flux density q [W.m^{-2}].

The simplest models of surface heat sources consider a constant heat flux on a rectangular or circular base (Figure 1).

$$q(x, y) = \frac{\Phi}{S} = q_{\max} \quad [\text{W.m}^{-2}] \quad (4)$$

while the affected area $S = a \times b$ and $S = \pi R_0^2$, respectively, and Φ is the effective power of a heat source [W].

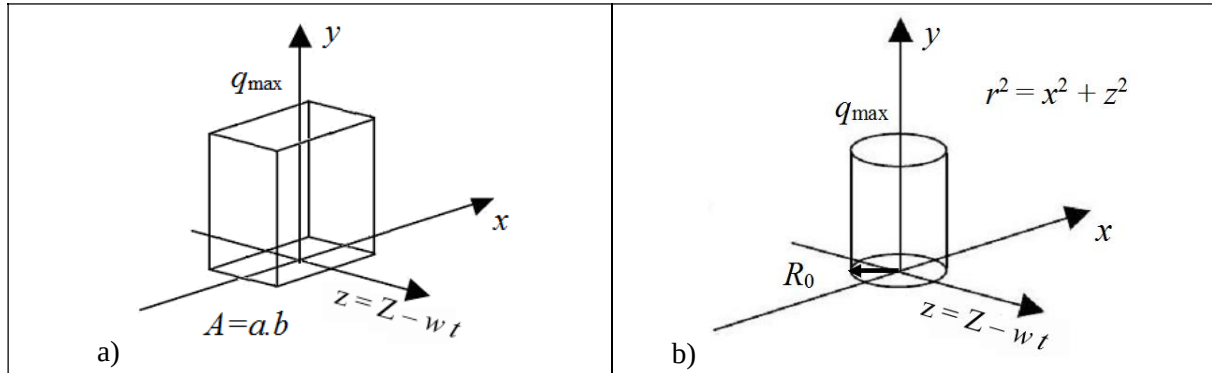


Figure 1. Surface heat source models with a constant heat flux on a) a rectangular base and b) a circular base.

In 1969, Pavelic [37] proposed a model of a surface heat source with a Gaussian normal distribution of heat flux density over a circular base. The distribution of the surface heat flux density on the surface of the welded material (Figure 2) depends on the radial position $r = (x^2 + z^2)^{1/2}$ from the heat source and is given by the relationship

$$q(x, y) = q_{\max} e^{-kr^2} = \frac{k\Phi}{\pi} e^{-kr^2}, \quad [\text{W.m}^{-2}] \quad (5)$$

where $q_{\max}$ is the maximum heat flux in the source centre for $r = 0$, Φ is the power of a heat source [W] and k is the heat flux concentration coefficient [m^{-2}].

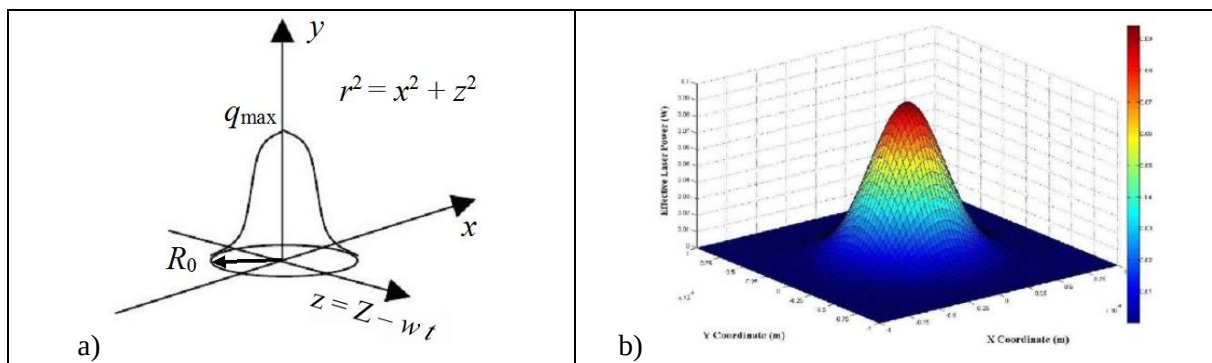


Figure 2. Surface heat source model with a Gaussian normal heat flux distribution a) a scheme and b) example of its application.

In general, surface heat source models are not suitable for numerical simulation of high-energy laser welding processes, and especially not for laser welding in the keyhole mode. However, they can be effectively used as an additional model to the volumetric heat source to better characterize the real effects of the applied heat source [38].

3.3. Volumetric heat source models

Volumetric heat source models are best suited to describe the nature of a moving laser energy source and to accurately capture the development of the weld pool, including the complex multi-physical phenomena occurring during a welding process. The term q_v [W.m^{-3}] in Eq. 1 is used to define the volumetric density of internal heat sources. The following section provides a brief characterization of the most frequently used models of volumetric heat sources, ranging from the simplest to the most complex.

3.3.1. Volumetric heat source models with a constant volumetric density of internal heat sources

These simple volumetric models assume a constant volumetric density of internal heat sources q_v in a defined volume, often with the shape of a block or cube (Figure 3a), or a hemisphere (Figure 3b). The volumetric density of internal heat sources is given by the relationship [39]

$$q_v(x, y, z) = \frac{\Phi}{V} = q_{\max} \quad [\text{W.m}^{-3}] \quad (6)$$

where Φ represents the effective power of a heat source [W], and $V = a \times b \times c$ for a block [m^3], and $V = (2\pi R_0^2)/3$ in the case of a hemisphere [m^3].

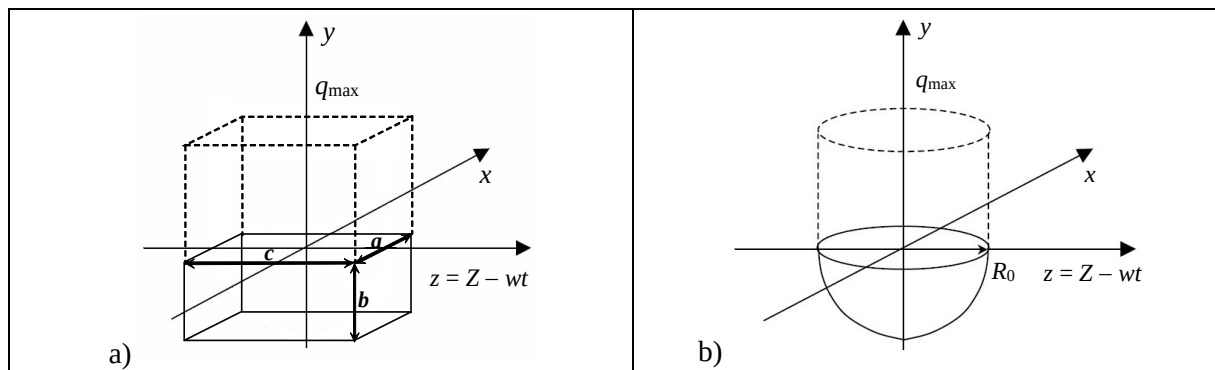


Figure 3. Volumetric heat source models with a constant volumetric density of internal heat sources in a) a block with dimension of $a \times b \times c$ and b) a hemisphere with a radius of R_0 .

The application of a constant volumetric density of heat sources to a volume with the shape of a truncated cone or pyramid can offer modifications to the presented models, making them more suitable for modelling laser beam welding processes [39-40]. However, the presented simple volumetric heat source models can be employed only for initial basic numerical simulations of laser welding processes, particularly for establishing the initial welding parameters.

3.3.2. Gaussian ellipsoidal volumetric heat source

The Gaussian ellipsoidal volumetric heat source (Figure 4) with Gaussian volumetric heat distribution is defined by three geometrical parameters - the ellipsoid half-axes a , b and c . The volumetric density of internal heat sources is described by a function [32, 39]

$$q_v(x, y, z) = \Phi \frac{6\sqrt{3}}{abc\pi^{3/2}} e^{-\frac{3x^2}{a^2}} e^{-\frac{3y^2}{b^2}} e^{-\frac{3z^2}{c^2}} \quad [\text{W.m}^{-3}] \quad (7)$$

where $V = a.b.c.\pi^{3/2}/(6\sqrt{3})$ is the volume of one half of the ellipsoid and the distribution function takes the form

$$f(x, y, z) = e^{-\frac{3x^2}{a^2}} e^{-\frac{3y^2}{b^2}} e^{-\frac{3z^2}{c^2}} \quad [-] \quad (8)$$

Application of the constant volumetric density of heat sources or Gaussian heat source to the volume with the shape of a truncated cone or pyramid represents the modifications of presented model more suitable for modeling of laser or electron beam welding processes [2].

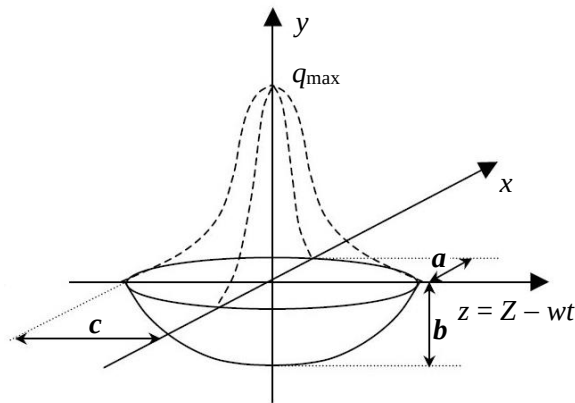


Figure 4. Gaussian ellipsoidal volumetric heat source with Gaussian heat distribution for modelling of volumetric internal heat sources to the volume of half-ellipsoid.

3.3.3. Goldak heat source model

The Goldak model or so called double ellipsoid heat source (Figure 5) describes the distribution of volumetric density of heat sources to a volume with the form of a double ellipsoid [41]. The double ellipsoid is characterized by six parameters: the half axes b and c represent the width and depth of the heat source, half axes c_1 and c_2 in the frontal and rear quadrants define its length, and parameters f_1 and f_2 designate the portion of the heat generated in the frontal and rear parts of the double ellipsoid.

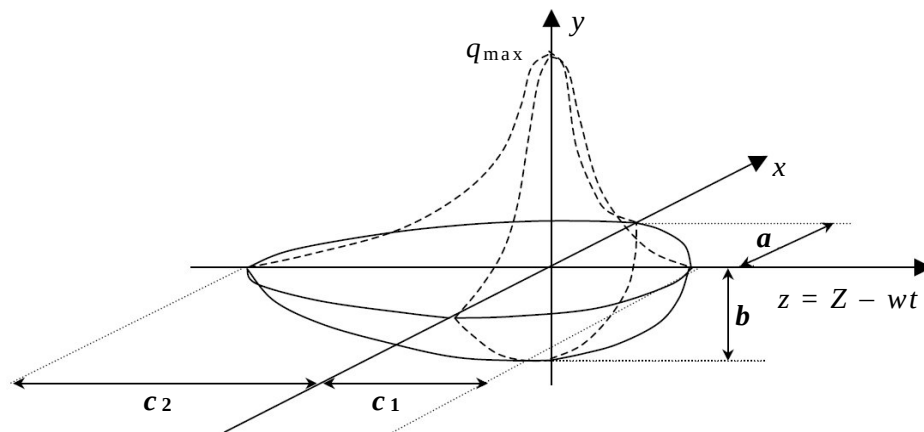


Figure 5. Goldak model for modelling of volumetric internal heat sources to the volume of double ellipsoid.

Volumetric density of heat sources in front and rear quadrants can be expressed as [41]

$$q_{v1}(x, y, z, t) = \frac{6\sqrt{3}f_1\Phi}{abc\pi^{3/2}} e^{-\frac{3x^2}{a^2}} e^{-\frac{3y^2}{b^2}} e^{-\frac{3(z+wt)^2}{c_1^2}} \quad [\text{W}\cdot\text{m}^{-3}] \quad (9)$$

$$q_{v2}(x,y,z,t) = \frac{6\sqrt{3}f_2\Phi}{abc_2\pi^{3/2}} e^{-\frac{3x^2}{a^2}} e^{-\frac{3y^2}{b^2}} e^{-\frac{3(z+wt)^2}{c_2^2}} \quad [\text{W.m}^{-3}] \quad (10)$$

$$f_1 = \frac{2c_1}{c_1 + c_2}, \quad f_2 = \frac{2c_2}{c_1 + c_2}, \quad f_1 + f_2 = 2 \quad [-] \quad (11)$$

Figure 6 illustrates the definition of the heat input into finite element model using different parameters of the Goldak model.

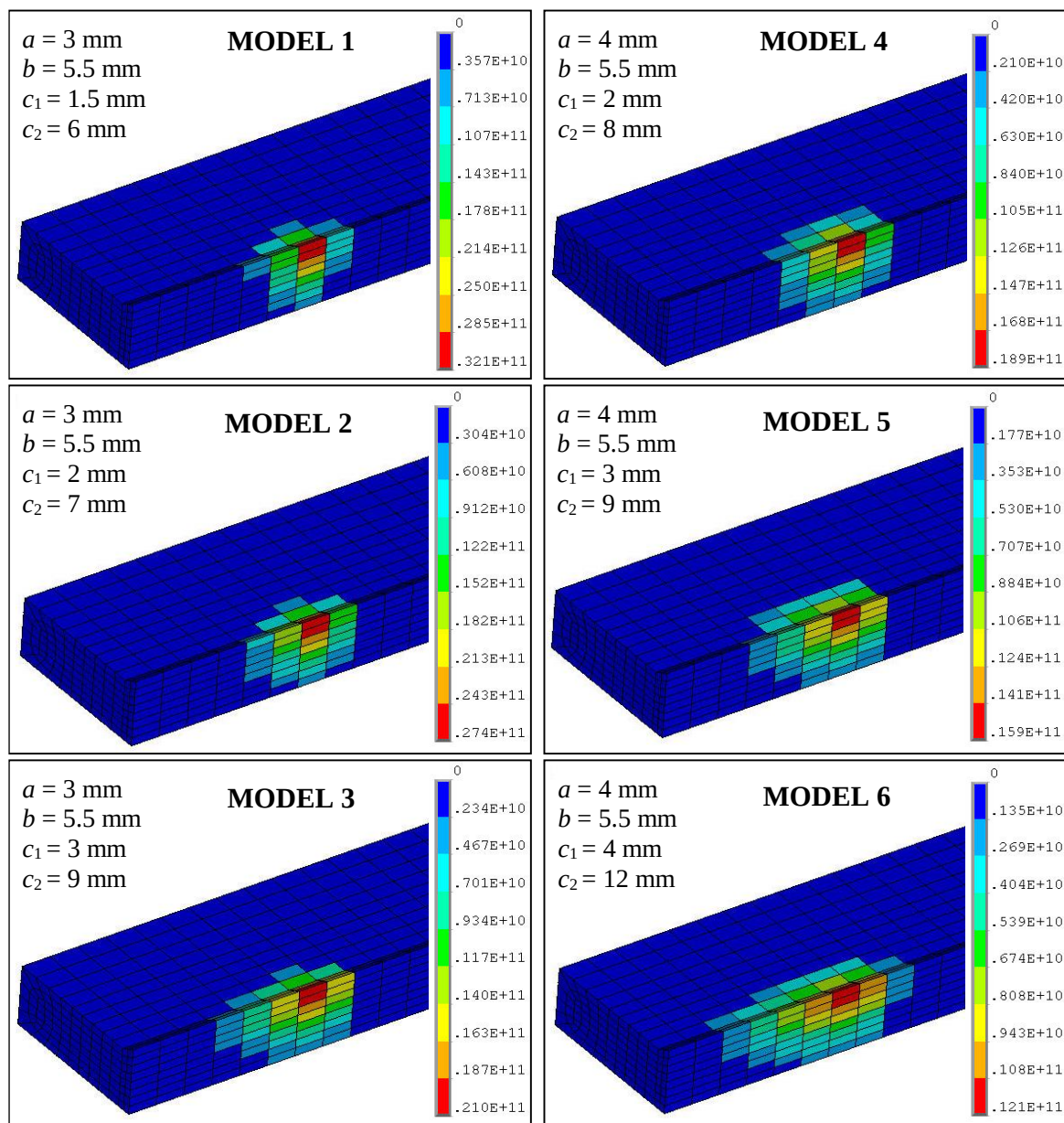


Figure 6. Application of the heat input q_v [W.m⁻³] to the FEM model using different parameters of the Goldak heat source model a , b , c_1 and c_2 (Eqs. 9–11).

The Goldak model can be used for numerical simulation of laser welding processes, but it is more suitable for simulations of arc welding processes. Generally, it is currently considered to be the most accurate but also the most demanding method for defining heat input to the weld in arc welding processes [16, 42].

It should be noted that the dimensions of the heat source are generally not known, and there is no direct link between welding process parameters and the dimensions of the heat source [41]. In most cases, the dimensions of the heat source are initially estimated for the simulation of a specific welding process and are subsequently modified and adjusted based on experimental results [42-46]. A portion of the heat supplied to both the front and back ellipsoids is determined by constants, f_1 and f_2 . These constants influence the energy distribution into the material, compensating for factors such as the inclination of the welding torch. A heat distribution ratio of 60:40 in favour of the front ellipsoid is mostly used [15, 47].

3.3.4. Conical heat source model

The conical heat source model, featuring a Gaussian distribution of volumetric internal heat sources (Figure 7), is commonly employed in numerical simulations of welding processes characterized by high energy density, including laser or electron beam welding processes.

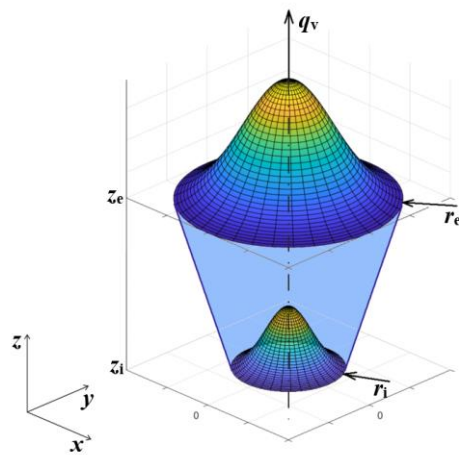


Figure 7. Conical heat source model.

This model is defined by the geometric parameters r_e , r_i , z_e and z_i (Figure 7) along with the welding speed w [m.s^{-1}], the total power of the source P and the efficiency η [-] of a heat source. The product of the total laser power and laser efficiency represents the effective power of a heat source $\Phi = \eta P$ [W]. The volumetric density of internal heat sources q_v [W.m^{-3}] can be expressed mathematically as [48-49]

$$q_v(x, y, z) = \frac{9\eta P e^3}{\pi(e^3 - 1)} \frac{1}{(z_e - z_i)(r_e^2 + r_e r_i + r_i^2)} \exp\left(-\frac{3[(x - wt)^2 + y^2]}{r_0^2(z)}\right) \quad [\text{W.m}^{-3}] \quad (12)$$

$$\text{where } r_0(z) = r_e + \frac{r_i - r_e}{z_i - z_e}(z - z_e) \quad [\text{m}] \quad (13)$$

The geometrical parameters r_e and r_i [m] are the surface radii in the planes, $z = z_e$ and $z = z_i$, respectively, and x , y , and z are the instantaneous spatial coordinates of the heat source. The parameter r_0 represents a distribution function whose value decreases linearly from the upper base of the truncated cone to its lower base.

In the case of partial penetration of the laser into the welded material, numerical simulation using a conical heat source model can provide realistic results. However, when full penetration is achieved, capturing the characteristics of weld joint expansion and the formation of the root region becomes challenging, particularly when welding thicker sheets [50, 51].

3.3.5. Cylindrical heat source model

According to the type of laser used for welding, various shapes of a weld pool can be developed. In the case of laser welding with a deep penetration, a narrow, cylindrical, or so-called bell-shaped weld joint can be formed. To describe this type of weld pool, a cylindrical volumetric heat source model, as illustrated in Figure 8 [44], can be employed.

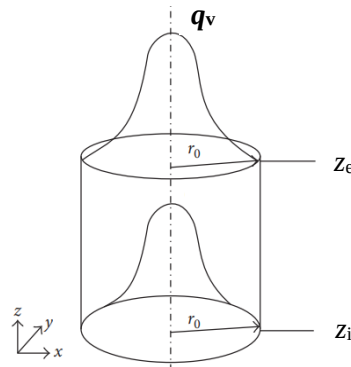


Figure 8. Cylindrical heat source model.

The volumetric density of internal heat sources q_v [$\text{W}\cdot\text{m}^{-3}$] can be calculated from the relationship

$$q_v(x, y, z) = \frac{2\eta Pe}{\pi(e^3 - 1)} \exp\left(-\frac{r^2}{r_0^2}\right), \quad [\text{W}\cdot\text{m}^{-3}] \quad (14)$$

where again P [W] is the total power of the source and η [-] is the efficiency of a heat source, and r_0 represents radius of the cylindrical heat source model.

3.3.6. Adaptive curve-rotated volumetric heat source model

The rotating volume model of the heat source [52], depicted in Figure 9, is primarily employed for modeling welding processes in the keyhole mode. This model better accommodates the complex distribution of the laser in the welded material than the volumetric heat sources introduced in the previous sections.

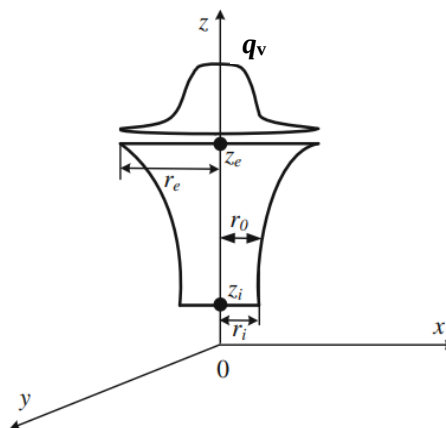


Figure 9. Adaptive curve-rotated volumetric heat source model [52].

The volumetric density of internal heat sources q_v [W.m^{-3}] for this model is described by following equations [52]

$$q_v(x, y, z) = \frac{3\eta P}{\pi(1-e^{-3})(E+F)} \times \left(\frac{1-\chi}{z_e - z_i} + \frac{\chi z_e - z_i}{z_e - z_i} \right) \exp\left(-\frac{3r^2}{r_0^2(z)}\right), \quad [\text{W.m}^{-3}] \quad (15)$$

$$r_0(z) = \frac{r_e - r_i}{z_e^2 - z_i^2} z^2 + \frac{r_i z_e^2 - r_e z_i^2}{z_e^2 - z_i^2}, \quad r = \sqrt{x^2 + y^2} \quad [\text{m}] \quad (16)$$

This volumetric heat source is defined by parameters r_e , r_i , z_e , and z_i according to Figure 9. The keyhole is characterized by the curve of the rotary volumetric heat source. Its height is equal to $(z_e - z_i)$, while the radius $r_0(z)$ tapers from top to bottom along the thickness of the welded materials. A proportion factor

$$\chi = \frac{q_m(0, z_i)}{q_m(0, z_e)} \quad (\chi \geq 1) \quad [-] \quad (17)$$

describes the difference between the peak power densities at the top ($z = z_e$) and bottom ($z = z_i$) surfaces of the domain $q_v(0, z_i)$ and $q_v(0, z_e)$ are the power densities at the central axis of bottom and top surfaces of the domain, respectively. The value $q_v(0, z)$ decreases linearly from $q_v(0, z_e)$ to $q_v(0, z_i)$.

$$E = \frac{1-\chi}{z_e - z_i} \left[\left(\frac{1}{p^2} \frac{z_e^6}{6} + \frac{s}{p} \frac{z_e^4}{2} + \frac{s^2}{2} z_e^2 \right) - \left(\frac{1}{p^2} \frac{z_i^6}{6} + \frac{s}{p} \frac{z_i^4}{2} + \frac{s^2}{2} z_i^2 \right) \right] \quad [\text{m}^3] \quad (18)$$

$$F = \frac{\chi z_e - z_i}{z_e - z_i} \left[\left(\frac{1}{p^2} \frac{z_e^5}{5} + 2 \frac{s}{p} \frac{z_e^3}{3} + s^2 z_e \right) - \left(\frac{1}{p^2} \frac{z_i^5}{5} + 2 \frac{s}{p} \frac{z_i^3}{3} + s^2 z_i \right) \right] \quad [\text{m}^3] \quad (19)$$

$$p = \frac{z_e^2 - z_i^2}{r_e - r_i}, \quad s = \frac{r_i z_e^2 - r_e z_i^2}{z_e^2 - z_i^2} \quad [\text{m}] \quad (20)$$

3.3.7. Variable Conical Profile and Full Variable Profile volumetric heat sources

Variable Conical Profile (VCP) and Full Variable Profile (FVP) volumetric heat sources (Figure 10) were developed primarily for numerical simulation of arc welding processes [16]. The concept behind these models is based on the necessity to consider the heat input distribution, as represented by the weld profile, during formulation.

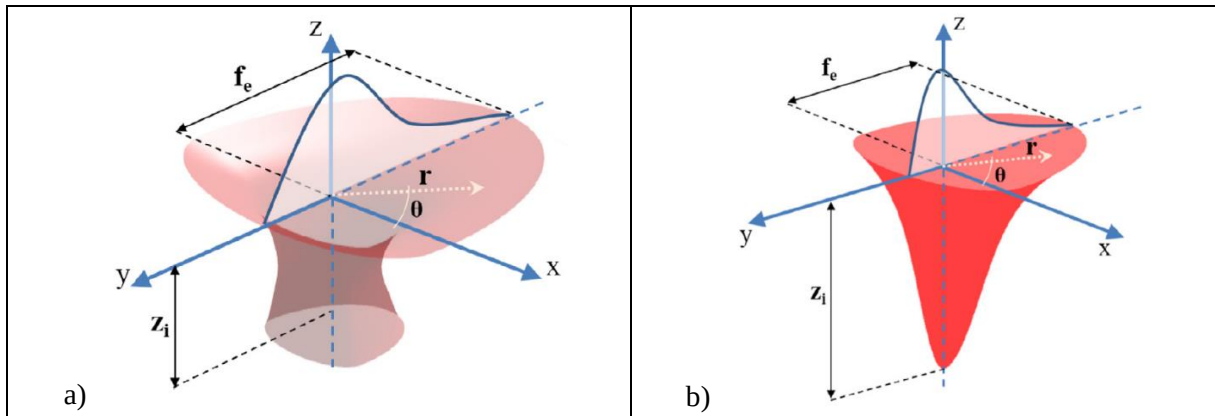


Figure 10. New volumetric heat sources developed by Farias et al [16]: a) Variable Conical Profile heat source and b) Full Variable Profile heat sources.

The shapes of these new volumetric heat sources are defined by the geometry of a fusion zone measured using weld macrographs. The primary distinction between these two heat sources lies in the decay of maximum power density through the thickness for the FVP source, contrasting with the VCP source where it remains constant. In contrast to the concept of hybrid heat sources composed of two different heat sources to adapt different types of welding processes, which will be introduced in the next section, the new heat sources have the advantage of being based on a single equation with a lower number of geometric parameters.

In case of Variable Conical Profile model (Figure 10a), the volumetric density of internal heat sources in front and rear parts, respectively, can be calculated from the equations [16]

$$q_{f_{VCP}}(r, z) = \frac{3\eta P}{\pi f_{size}^2 (1 - e^{-3}) \int_{z_1}^{z_e} p_f^2 dz} \exp\left(-\frac{3r^2}{r_F^2}\right) \quad [W.m^{-3}] \quad (21)$$

$$q_{r_{VCP}}(r, \theta, z) = \frac{3\eta P}{\pi f_{size}^2 f_{ellipse} (1 - e^{-3}) \int_{z_1}^{z_e} p_f^2 dz} \exp\left(-\frac{3r^2}{r_R^2}\right) \quad [W.m^{-3}] \quad (22)$$

while $f_{ellipse}$ is ellipse factor, which determines the elongation of the rear quadrant of the heat source in relation to the front quadrant. This represents a characteristic of the weld pool during the welding process. The f_{size} is size factor, which multiplies the basic heat source volume, so it allows to adjust the power density. The term f_f represents the polynomial function that describes the weld profile in the thickness direction.

In the case of Full Variable Profile heat source model (Figure 10b) the power density decreases gradually from the center of the volume in three orthogonal directions, including the direction along the plate thickness. This behaviour is attained by introducing an additional term into the presented equations for the VCP model [16], resulting in an exponential decay in the thickness direction on the radial coordinate r_F and r_R for frontal and rear part of the FVP heat source model, respectively.

$$q_{f_{FVP}}(r, z) = \frac{3\eta P}{\pi f_{size}^2 (1 - e^{-3}) \int_{z_1}^{z_e} p_f^2 \exp\left(-\frac{3z^2}{z_i^2}\right) dz} \exp\left(-\frac{3r^2}{z_F^2} - \frac{3z^2}{z_i^2}\right) \quad [W.m^{-3}] \quad (23)$$

$$q_{r_{FVP}}(r, \theta, z) = \frac{3\eta P}{\pi f_{size}^2 f_{ellipse} (1 - e^{-3}) \int_{z_1}^{z_e} p_f^2 \exp\left(-\frac{3z^2}{z_i^2}\right) dz} \exp\left(-\frac{3r^2}{z_R^2} - \frac{3z^2}{z_i^2}\right) \quad [W.m^{-3}] \quad (24)$$

3.3.8. Hybrid volumetric heat sources

As previously mentioned, hybrid volumetric heat sources are composed of two or more different heat sources to better describe the complex physics behind different types of welding processes and to adjust the parameters of the resulting model so that the obtained results correspond to the real measured shape of the weld pool or weld macrographs.

In numerical simulation of laser welding processes, the double-conical volumetric heat source model (Figure 11) is the most frequently employed hybrid heat source model. The distribution of internal heat sources in the case of the double-conical volumetric heat source model is described by the following equations [53]

$$q_{v1}(x, y, z) = \frac{9\eta Pe^3}{\pi(e^3 - 1)(z_i - z_b)(r_i^2 + r_i r_b + r_b^2)} \exp\left(-\frac{3[(x - wt)^2 + y^2]}{r_i^2}\right) \quad [W.m^{-3}] \quad (25)$$

$$q_{v2}(x, y, z) = \frac{9\eta Pe^3}{\pi(e^3 - 1)(z_t - z_i)(r_t^2 + r_t r_i + r_i^2)} \exp\left(-\frac{3[(x - wt)^2 + y^2]}{r_2^2}\right) \quad [\text{W}\cdot\text{m}^{-3}] \quad (26)$$

$$r_1 = r_i - (r_i - r_b) \frac{z_i - z}{z_i - z_b} \quad [\text{m}] \quad (27)$$

$$r_2 = r_t - (r_t - r_i) \frac{z_t - z}{z_t - z_i} \quad [\text{m}] \quad (28)$$

The parameters used have a similar meaning to those in the simple conical model and can be identified from Figure 11a. Figure 11b illustrates some simple modifications of double-conical heat source with reverse and direct cone configurations.

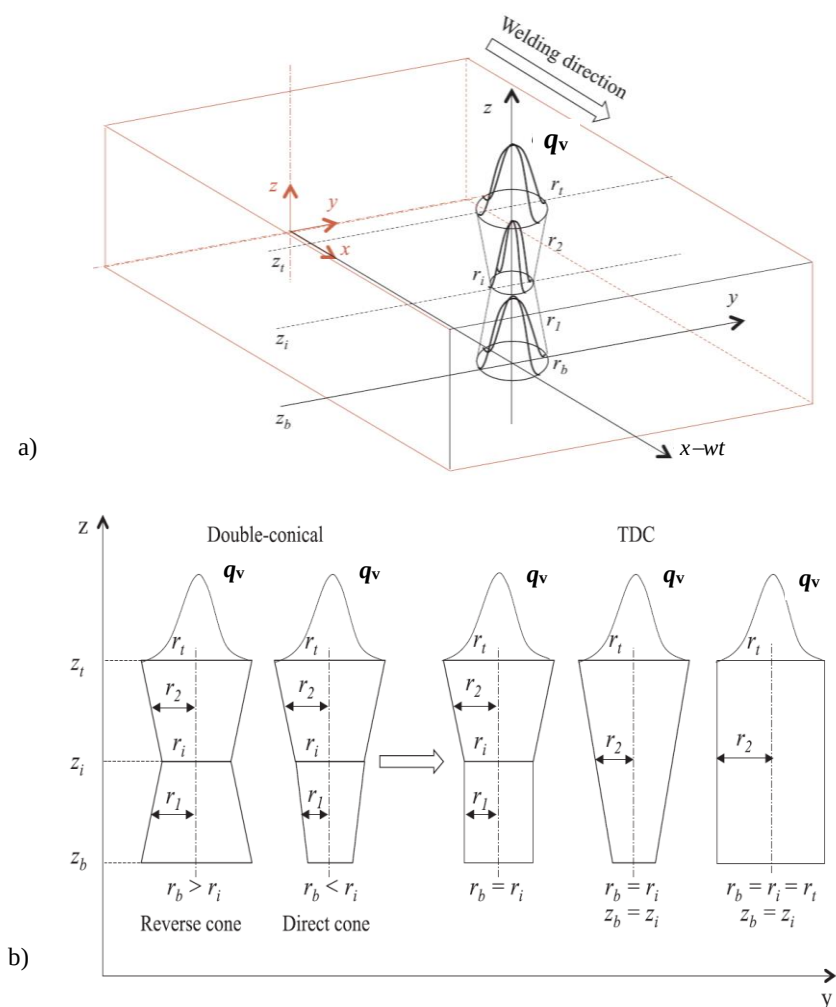


Figure 11. Double-conical volumetric heat source model a) a scheme of the model with reverse cone configuration and b) simple modifications of double-conical heat source with reverse and direct cone configurations [51].

In a similar way, it is possible to combine different types of heat source models for a more accurate description of the resulting weld joint, based on the assumed shape and the welding method employed. Figures 12 and 13 illustrate some examples of applied combinations of volumetric heat sources used to model various types of welding processes, as documented in the literature.

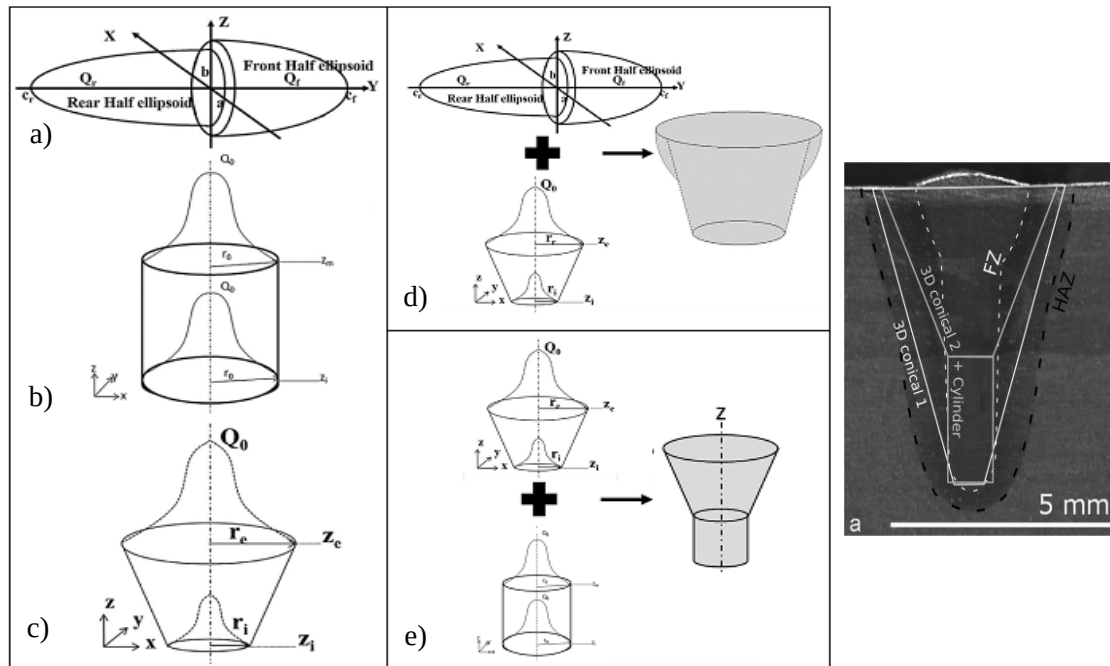


Figure 12. Selected volumetric heat source models and their combinations a) Goldak model, b) cylindrical model of the heat source, c) conical model of the heat source, d) combination of 3D conical and Goldak models, e) combination of conical and cylindrical model of the heat source [44, 14].

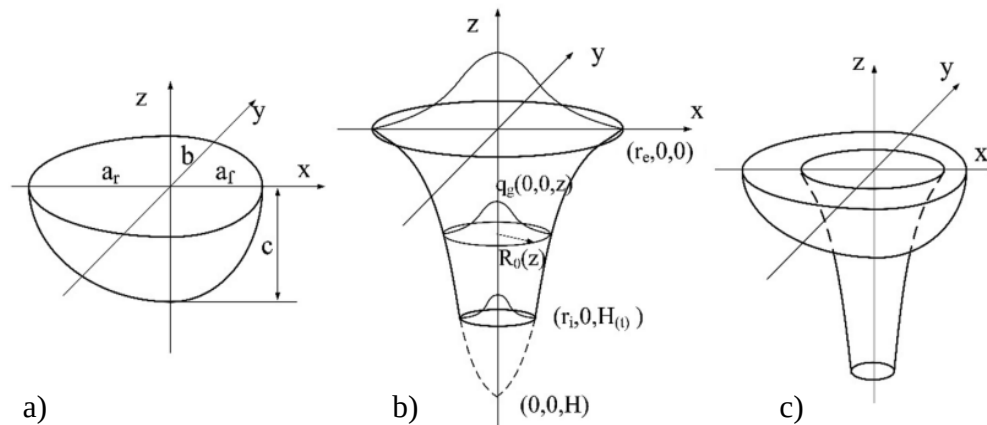


Figure 13. Combination of Goldak and adaptive curve-rotated volumetric heat source models [54].

4. Influence of parameters of the conical heat source model on weld pool characteristics

In the next part, the influence of the geometric characteristics of the conical heat source model on the calculated dimensions and maximum temperatures of the weld pool will be demonstrated using the example of laser welding of steel sheets with a thickness of 4 mm. Numerical simulation of laser welding of sheets made of S650MC steel by a TruDisk 4002 disk laser with a maximum power of 2 kW and a laser fiber diameter of $\varnothing 400 \mu\text{m}$ was performed in the program code ANSYS.

4.1. Simulation model

The geometric model in the shape of a cuboid had dimensions of $40 \times 35 \times 4 \text{ mm}$, while the symmetry of the weld joint was taken into account. The finite element mesh (Figure 14) was generated using

3D-element SOLID70 [55]. The elements in the densest part of the mesh in the weld zone have dimensions of $0.1 \times 0.2 \times 0.2$ mm.

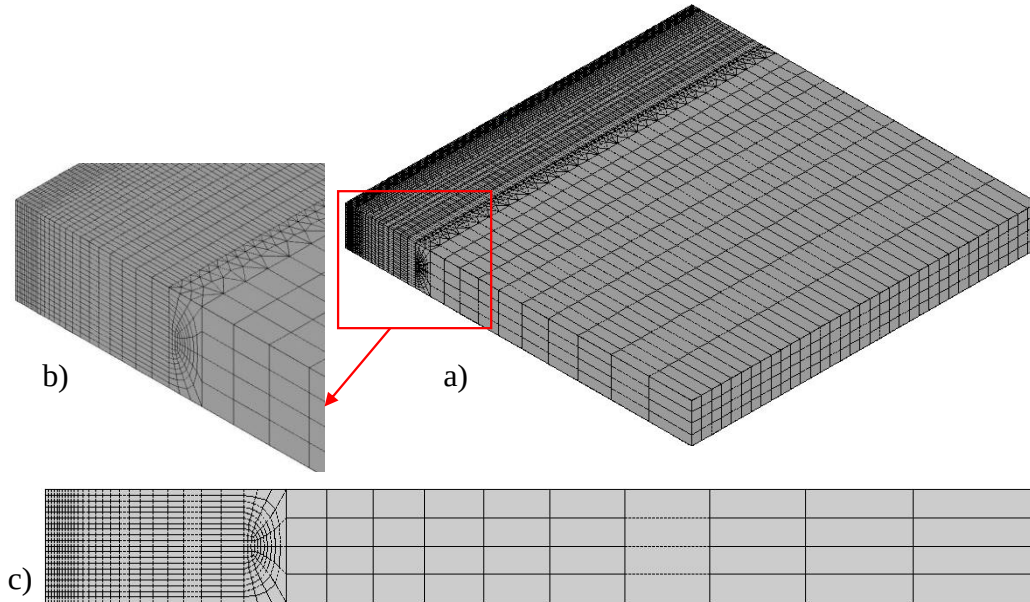


Figure 14. Finite element model (a) with detail in the weld area (b) and front view (c).

The S650MC steel used is characterized by good weldability and formability, which allows for its quick and efficient processing. The material properties of S650MC steel were calculated based on its chemical composition (Table 1) using JMatPro software [56]. Figure 15 shows the dependencies of thermal conductivity, density, and specific heat capacity on temperature. The calculated solidus temperature is $T_s = 1414$ °C and the liquidus temperature is $T_L = 1501$ °C.

Table 1. Chemical composition of S650MC steel.

Element	S (max.)	Al (min.)	Nb (max.)	V (max.)	Ti (max.)	C (max.)	Si (max.)	Mn (max.)	P (max.)
[wt. %]	0.010	0.015	0.09	0.20	0.15	0.12	0.21	2.00	0.025

To define the loads in the form of heat input during laser welding, the conical volumetric model of a heat source was used. The laser power was assumed to be 1200 W and 1500 W at an efficiency of 63.5 %. The welding speed was 8 mm.s^{-1} .

The parameters of the conical volumetric heat source were supposed to be from the intervals:

- radius of the upper base of the truncated cone r_e : 0.3 mm – 1.6 mm
- radius of the lower base of the truncated cone r_i : 0.2 mm – 1.0 mm
- height of the truncated cone h : constant value of 4 mm.

Figure 16 illustrates the volumetric density distribution of internal heat sources for selected parameters of the conical heat source model.

The initial temperature of the welded plates was assumed to be 20 °C. Heat dissipation from the surface of the welded plates into the surrounding air was in this case neglected due to the short duration of the entire process.

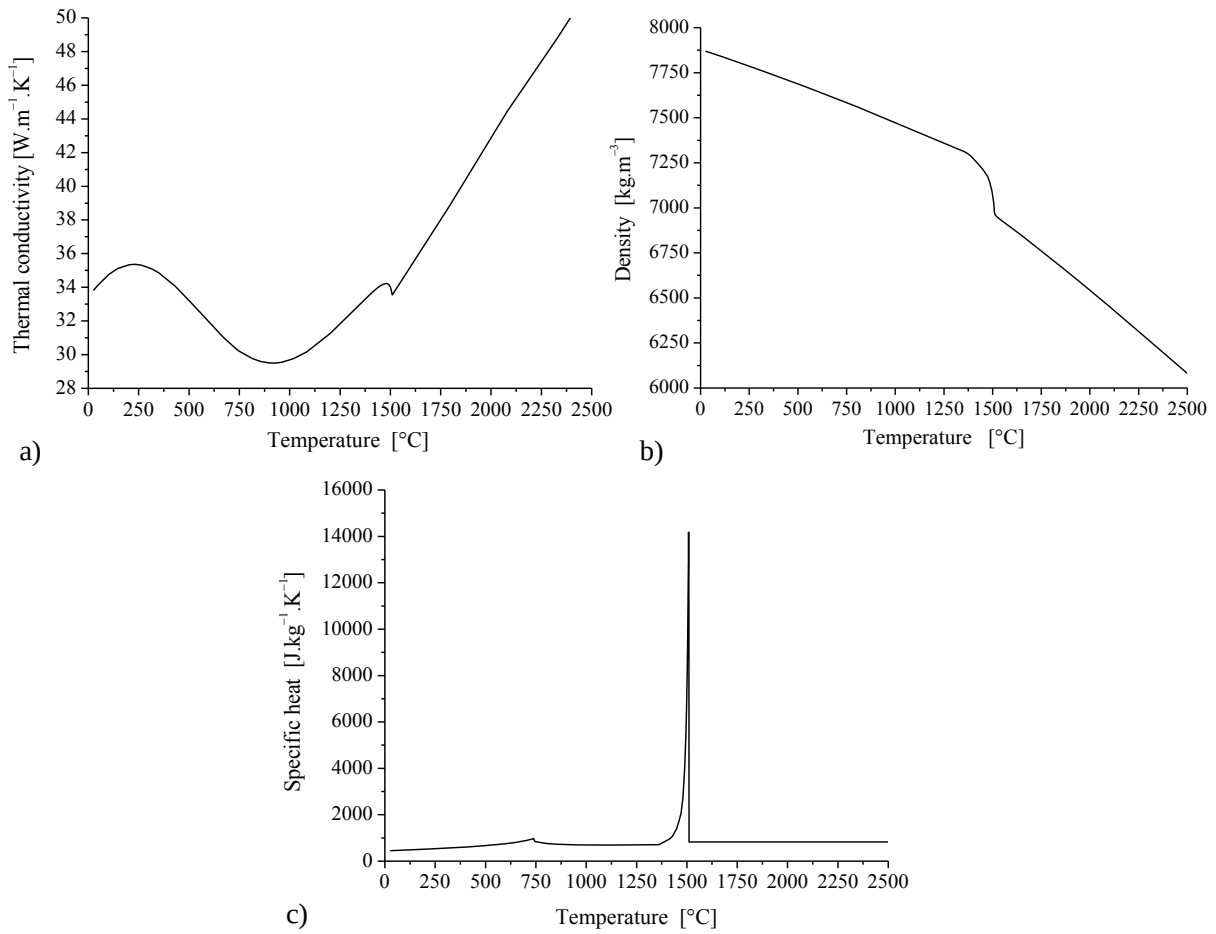


Figure 15. Thermal properties of the S650MC steel in the dependence on temperature
a) thermal conductivity b) density and c) specific heat.

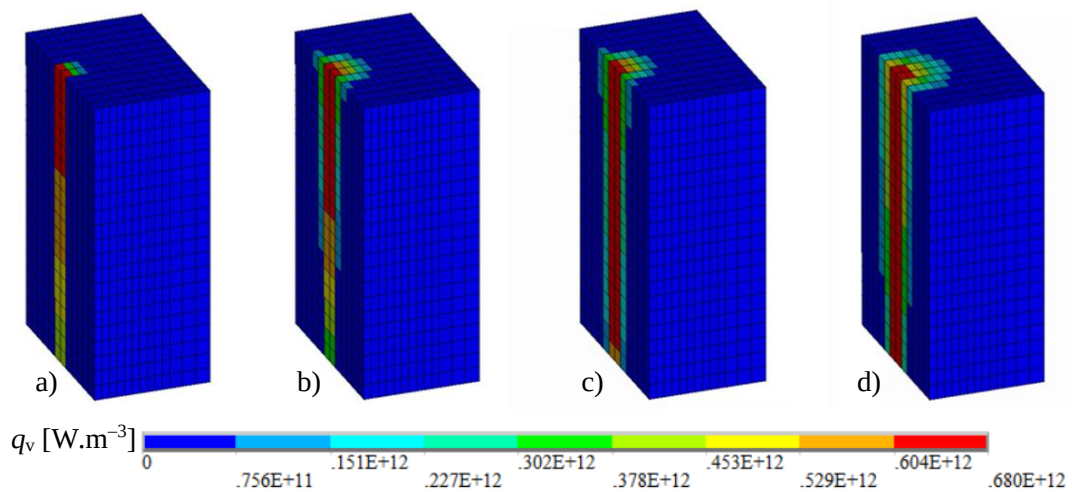


Figure 16. Volumetric density distribution of internal heat sources q_v $[W.m^{-3}]$ for selected parameters of the conical heat source model a) $r_e = 0.3$ mm, $r_i = 0.2$ mm, b) $r_e = 0.6$ mm, $r_i = 0.2$ mm, c) $r_e = 0.6$ mm, $r_i = 0.4$ mm a d) $r_e = 0.8$ mm, $r_i = 0.5$ mm.

4.2. Influence of parameters of the conical heat source model on the geometric characteristics and maximum temperatures of a weld pool

Tables 2 and 3 summarize the computed shapes of weld pool for different values of r_e and constant value of $r_i = 0.8$ mm at 1500 W laser power and $r_i = 0.2$ mm at 1200 W laser power, respectively. The temperature range corresponds to the temperature interval between solidus and liquidus temperatures for S650MC steel. The zones inside have temperatures higher than the liquidus temperature, indicating that they represent the fused zones.

As it can be seen in Table 3, complete remelting of the material over the entire thickness at laser power of 1200 W occurs only for $r_e = 0.3$ mm. The width of the weld pool increases with the growing size of r_e , but the size of the molten area in the cross-section decreases.

Table 2. Weld pool shapes for selected values of r_e and $r_i = 0.5$ mm at a laser power of 1500 W.

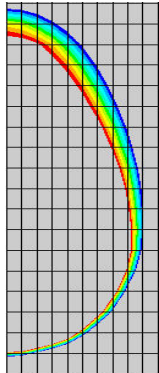
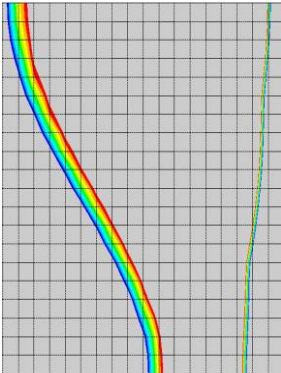
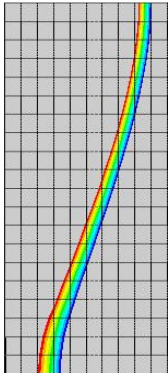
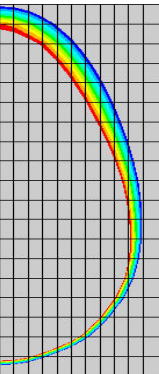
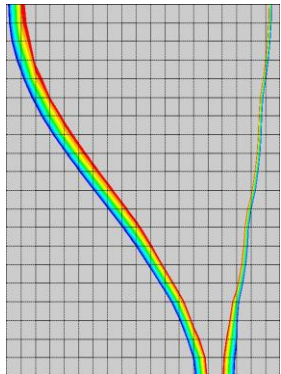
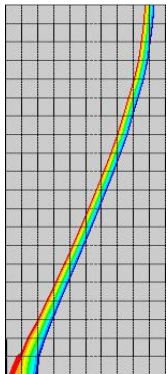
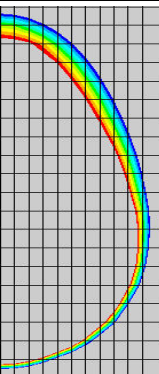
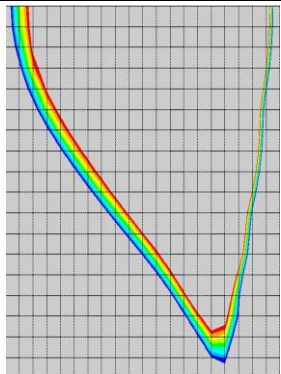
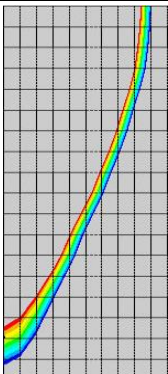
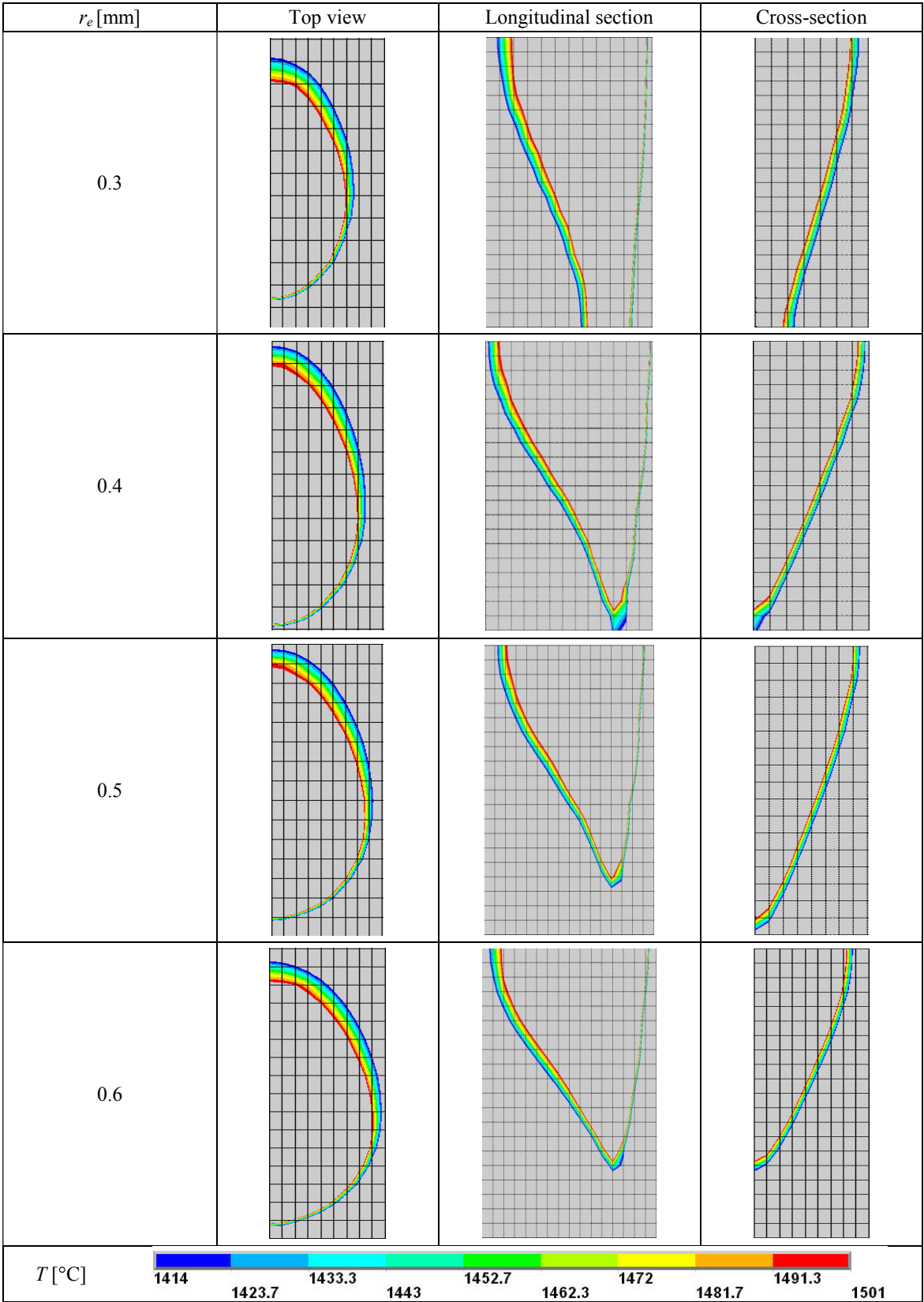
r_e [mm]	Top view	Longitudinal section	Cross-section
0.8			
1			
1.2			
T [°C]  1414 1423.7 1433.3 1443 1452.7 1462.3 1472 1481.7 1491.3 1501			

Table 3. Weld pool shapes for selected values of r_e and $r_i = 0.2$ mm at a laser power of 1200 W.



Figures 17-20 show the dependences of width, length and depth of the weld pool on the upper radius of conical heat source r_e for selected bottom radius of conical heat source r_i and laser power of 1200 W and 1500 W.

As the radius of the upper base of the truncated cone r_e increases, with a constant value for r_i , the width and length of the weld pool nonlinearly increase, while of the weld pool decreases. When considering a power of 1200 W, the sheets are fully remelted when the condition $r_e \leq r_i + 0.2$ mm is met.

With a constant value for r_i , the maximum temperature of the weld pool increases as the radius r_e rises until complete penetration of the materials is achieved.

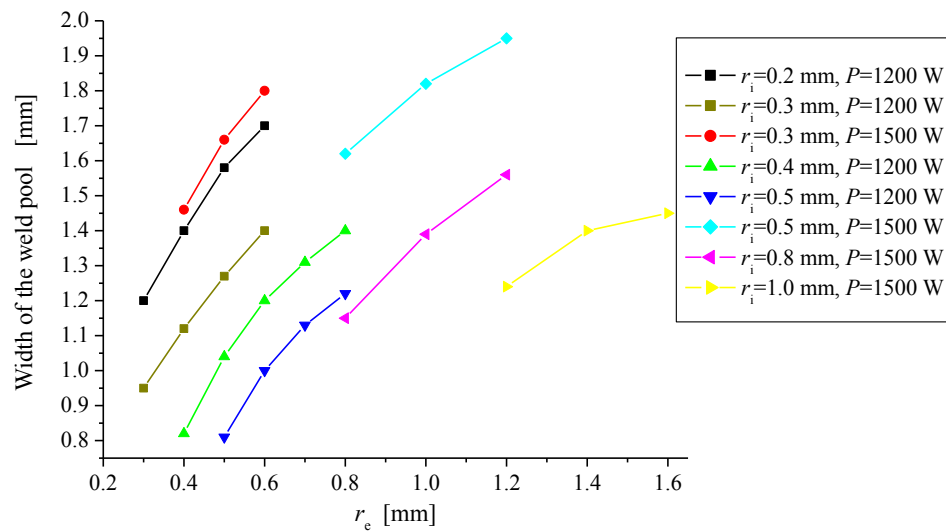


Figure 17. Effect of geometrical characteristics of the conical volumetric heat source model on the computed width of the weld pool for laser powers of 1200 W and 1500 W.

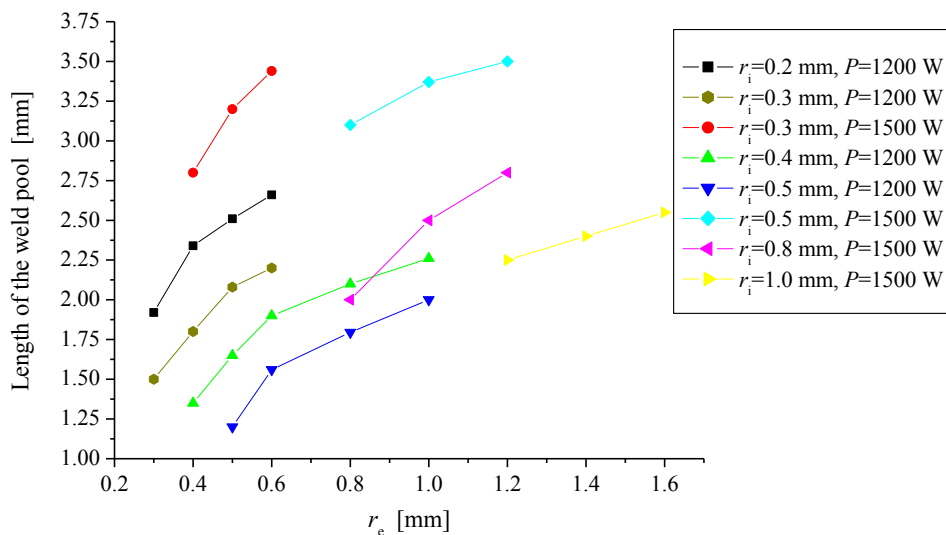


Figure 18. Effect of geometrical characteristics of the conical volumetric heat source model on the computed length of the weld pool for laser powers of 1200 W and 1500 W.

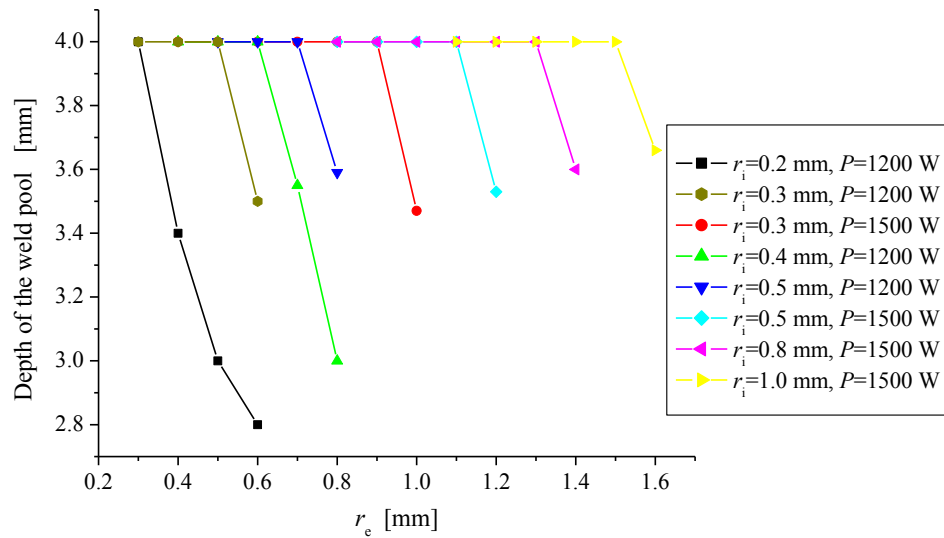


Figure 19. Effect of geometrical characteristics of the conical volumetric heat source model on the computed depth of the weld pool for laser powers of 1200 W and 1500 W.

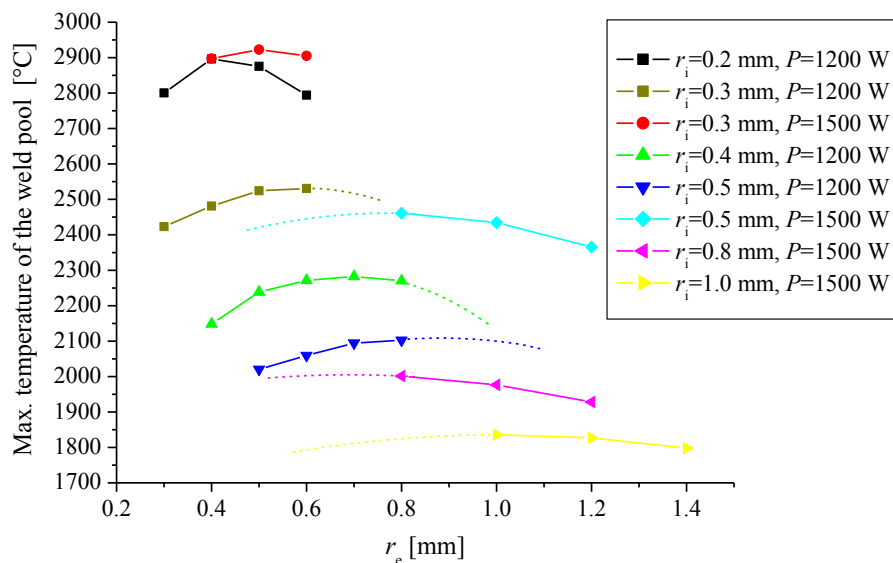


Figure 20. Effect of geometrical characteristics of the conical volumetric heat source model on the computed maximum temperature of the weld pool for laser powers of 1200 W and 1500 W.

As evident from the obtained results of numerical simulations, the input geometrical parameters significantly affect the computed shape and dimensions of the weld pool. For this reason, it is recommended to validate computed results through experimental temperature measurements and compare the computed dimensions of the fusion zone with weld macrographs. Finally, different calibration methods of applied heat source model can be employed to ensure accurate results of numerical simulations that closely correspond to real [19, 22, 57-58] welding conditions, capturing the real macrostructure and properties of the produced weld joint.

Conclusions

The paper offers a brief overview of various heat source models designed for the numerical simulation of fusion welding processes, with a specific emphasis on laser welding. Moreover, the influence of parameters of the conical volumetric heat source model on the characteristics of the weld pool and maximum temperatures in fusion zone in laser welding is illustrated using an example of numerical simulation for the production of a butt weld joint of sheets made of S650MC steel.

It must be noted, that most of the presented heat source models are not especially tailored for advanced multi-physical numerical simulations of laser welding in the keyhole regime. Such models and simulations typically encompass a range of phenomena, including light-matter interaction, conduction, melting, melt convection in the fusion zone, vaporization and effects of vapor pressure, plasma generation, surface deformation, and others. A comprehensive literature review addressing numerical simulation of keyhole laser welding is available e.g. in [2].

Another note concerns the validation and calibration of heat sources for numerical simulation of welding processes. These processes are essential steps in the development and application of numerical heat source models. Validation ensures that the simulation model accurately represents the physical processes occurring during real welding, that the results provided can be considered credible, and that the model can be used for predictive purposes. The purpose of the calibration process is to adjust the parameters of the heat source model used to improve its accuracy and alignment with reality. Obviously, the parameters of a heat source model are “fine-tuned” based on experimental data (primarily temperature measurement or a comparison with macrostructure of weld joint) until the simulated results closely match the observed outcomes. Calibration significantly enhances the predictive capability of the simulation model, making it more reliable for various welding applications. Further insights into this matter are provided in [27].

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